

Project Proposal

1. Project Title: GelDoc? — Agarose Gel Diagnostics & Wet-Lab Protocol Troubleshooter

- **Project Manager (PM):** Isabelle Li

2. Overview / Summary

In molecular biology, agarose gel electrophoresis is the universal diagnostic test for DNA extraction, PCR amplification, and plasmid cloning. However, gels frequently produce ambiguous artifacts—such as faint target bands, primer-dimer smears running at the bottom, or non-specific off-target products. In complex biological systems like the green hydra endosymbiosis (*Hydra viridissima* and *Chlorella*) project I did over the summer, researchers must navigate five distinct co-existing genomes (host nuclear/mitochondrial and algal nuclear/mitochondrial/chloroplast DNA). Differentiating a true 18S or 5S target band from non-specific host amplification or primer dimers often takes days of manual ladder measuring and protocol guesswork.

Accomplished to Date (Previous Semester / Baseline):

I have engineered and deployed a functional web-based bioinformatics application built with Vue 3, Vuetify 3, Vite, and FastAPI. The platform successfully ingests .ab1 Sanger sequencing trace files, performs pairwise and multiple sequence alignments, and generates consensus sequences.

Current Proposed Extension: GelDoc would expand the existing architecture from a downstream Sanger viewer into an end-to-end wet-lab diagnostic platform. The new tool pairs Computer Vision (YOLO/OpenCV) with Generative AI reasoning (Gemma 4 / RAG). Users upload an unannotated gel photograph alongside experimental parameters; the system automatically detects lanes and bands, calculates base-pair sizes against a standard ladder, and utilizes an LLM reasoning agent grounded in lab protocols to prescribe concrete wet-lab fixes (e.g., adjusting annealing temperatures, identifying expired ligase, or recommending 5' M13 primer adapters).

3. Goals

- **Automated Gel Feature Extraction:** Train a lightweight YOLO model to detect wells, sample lanes, DNA ladders, resolved bands, and low-molecular-weight smears from raw smartphone or camera photos.
- **Deterministic Base-Pair Sizing:** Implement an image-processing module to compute migration distances and map band positions to exact base-pair sizes using ladder comparison.

- **Protocol & Failure Diagnostics via GenAI:** Connect a multimodal LLM to a database containing standard cloning guides (e.g., pGEM-T Easy manuals, PCR troubleshooting guides). The agent will generate actionable recommendations when bands fail, smear, or amplify off-target.

4. Tech Stack

- **Frontend:** React/Vue 3, HTML5 Canvas / SVG (for dynamic bounding box overlays and trace inspection).
- **Backend:** FastAPI (Python 3.11+).
- **Computer Vision & Sizing:** YOLO (Ultralytics YOLOv8/YOLO11), OpenCV, NumPy, SciPy (curve fitting).
- **Bioinformatics & Parsers:** Biopython (Bio.SeqIO.ABioIO for trace extraction, Bio.Seq for sequence validation).
- **Generative AI & Knowledge Retrieval**

5. Member Requirements

- **Recruitment Target:** 2–4 members
 - **1–2 ML:** Focus on YOLO dataset labeling, training/fine-tuning gel band detection, and implementing the OpenCV migration distance curve-fitting algorithm.
 - **1 Backend:** Focus on FastAPI endpoint design
 - **1 Frontend:** Focus on building an image uploading interface, interactive canvas overlays for detected bands, and integrating API responses with the existing consensus viewer.
- **Time Commitment:** 6–8 hours per week?
- **Prerequisites:**
 - Proficiency in Python (NumPy, basic API concepts) or modern JavaScript/TypeScript (Vue/React).
 - Familiarity with Git workflows.
 - Interest in bioinformatics, machine learning, or computer vision (wet-lab molecular biology background is a plus, but not needed).

6. Timeline (hopefully)

Weeks 1-3

- Onboard team to repo, set up environments
- Collect, annotate, and label gel datasets
- Train initial machine baseline and image uploading platform

Weeks 4-7

- Integrate ladder sizing for base-pair sizes
- Integrate lab manual guides

Weeks 8-10

- Connect ends and diagnostics
- Build interactive canvas for ladder display
- Validate edge cases

Other weeks?

- Polish and refine for presentation